

Variational Principles and Discretization in Stochastic Thermodynamics

Xu Zixuan

College of Computing and Data Science

François Gay-Balmaz

School of Physical and Mathematical Sciences

Abstract. Variational formulations of nonequilibrium thermodynamics extend Hamilton's principle to systems with irreversible processes such as friction and heat conduction. While deterministic structure-preserving discretization exists, the corresponding methods for stochastic thermodynamic systems that preserve both energy balance and entropy production remain underdeveloped. Standard stochastic integrators may fail to preserve the entropy production structure and fluctuation-dissipation relation. In this paper, we develop thermodynamically consistent numerical schemes for stochastic systems formulated in the Stratonovich framework.

We consider finite-dimensional thermodynamic systems derived from a variational principle and extend them to stochastic thermodynamics through Langevin-type forces satisfying fluctuation–dissipation relations. Deterministic systems are discretized using discrete gradient methods that guarantee exact energy balance and non-negative entropy production. We then construct discretization schemes by extending discrete gradient methods and stochastic integrators to systems with noise. The proposed algorithms are then applied to stochastic models. We show that deterministic discrete gradient schemes conserve total energy exactly in isolated systems. In the stochastic case, we prove discrete energy balance, and will further discuss entropy production, and recovery of equilibrium distributions consistent with the Einstein fluctuation–dissipation relation. We derive the one-step transition kernel, compare forward and reversed path probabilities, and use their ratio to define the discrete medium entropy production, thereby deriving a discrete general fluctuation-dissipation relation.

Keywords: Stochastic Thermodynamics, Discrete Gradient Methods, Structure Preserving Discretization

1 INTRODUCTION

Classical thermodynamics provides a foundation for energy and entropy, where fluctuations are usually negligible, and the evolution of a system can be described by deterministic equations. However, in mesoscopic systems, these fluctuations become significant. These fluctuations between the system and the en-

vironment will not only be affected by deterministic forces, but also by stochastic forces.

Stochastic thermodynamics will provide a method for studying these mesoscopic systems where thermal fluctuations are relevant by extending principles such as entropy production, and fluctuation–dissipation relation to individual random trajectories as well as at the ensemble scale [8, 9].

We explore how stochastic thermodynamic systems can be discretized in a way that preserves their structure. While standard numerical methods are mainly used for accuracy or convergence, they fail to preserve physical properties when applied to small thermodynamic systems. In particular, they may not preserve entropy production and fluctuation-dissipation balance. This motivates the use of discrete gradient schemes or geometric integrators that conserve the Hamiltonian [4].

The goal of this paper is to derive a structure-preserving stochastic discretization that is compatible with the fluctuation-dissipation relation. First, we use the discrete gradient schemes for stochastic thermodynamic systems and ensure that the first law is verified using the chain rule. Second, to derive the discrete form of entropy production, we will formulate the one-step forward and backward transition kernel. Third, we will obtain the discrete medium entropy production from the forward-to-reversed transition probability ratio. Finally, we derive a general entropy-dependent discrete fluctuation-dissipation relation.

The paper is organized as follows. Section 2 provides a brief review of stochastic differential equations, including stochastic calculus and the Wiener process. Section 3 introduces the variational principles for stochastic thermodynamics. Section 4 reviews the deterministic thermodynamic discretization. Section 5 reviews the stochastic thermodynamic discretization. In Section 6, we derive the forward and reversed kernels, formulate the discrete entropy production, and derive the discrete fluctuation-dissipation relation. The final sec-

tion summarizes the results and discusses future work and improvements.

2 REVIEW OF STOCHASTIC DIFFERENTIAL EQUATIONS

The stochastic thermodynamic systems used in this paper are described by stochastic differential equations. Formally, one often writes a stochastic differential equation in the form

$$\dot{x}(t) = A(x(t)) + B(x(t)) \eta(t), \quad (1)$$

where A is the deterministic drift, B is the noise amplitude, and $\eta(t)$ is Gaussian white noise. While white noise is not an ordinary function of time, we can see it as the time derivative of a Wiener process:

$$\eta(t) = \frac{dW(t)}{dt}.$$

Thus, the stochastic differential equation can be written as

$$dx(t) = A(x(t)) dt + B(x(t)) \overset{\alpha}{\circ} dW(t),$$

where the symbol $\overset{\alpha}{\circ}$ specifies the stochastic interpretation of the integral. The parameter $\alpha \in [0, 1]$ determines where the integrand is evaluated inside each time interval. When $\alpha = 0$, it corresponds to the Itô convention, and $\alpha = \frac{1}{2}$ corresponds to the Stratonovich convention [5, 6]. Throughout this paper, we will use the Stratonovich convention.

We define the Wiener process as

$$\mathbb{E}[W(t)] = 0, \quad \mathbb{E}[W(t)W(s)] = 2 \min(t, s).$$

such that it satisfies

$$\Delta W_n \sim \mathcal{N}(0, 2h).$$

The stochastic integral in (1) can be defined as a limit of discrete sums:

$$\begin{aligned} & \int_0^t B(x(s)) \overset{\alpha}{\circ} dW(s) \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n B(x_{i-1} + \alpha(x_i - x_{i-1})) \Delta W_i. \end{aligned}$$

The choice of α determines the stochastic convention. The Stratonovich convention corresponds to $\alpha = \frac{1}{2}$, where the integrand is evaluated at the midpoint:

$$\int_0^t B(x(s)) \circ dW(s) = \lim_{n \rightarrow \infty} \sum_{i=1}^n B\left(\frac{x_i + x_{i-1}}{2}\right) \Delta W_i.$$

2.1 STOCHASTIC CALCULUS

Let $x(t)$ be the solution to a stochastic differential equation:

$$dx(t) = A(x(t)) dt + B(x(t)) \overset{\alpha}{\circ} dW(t),$$

and let $f(x)$ be a smooth function of $x(t)$. We want to compute the stochastic differential of $f(x(t))$. Since dW is of order $\sqrt{dt}$, a Taylor expansion of $f(x + dx)$ must retain terms of order $(dx)^2$. Applying a generalized version of the chain rule [6] yields

$$\frac{d}{dt} f(x(t)) = f'(x(t)) \dot{x}(t) + D(1 - 2\alpha) f''(x(t)) B(x(t))^2. \quad (2)$$

where D is such that

$$\langle \eta(t) \eta(s) \rangle = 2D \delta(t - s).$$

For the rest of the paper we take $D = 1$.

For the Stratonovich convention $\alpha = \frac{1}{2}$, the correction term in (2) vanishes

$$\frac{d}{dt} f(x(t)) = f'(x(t)) \dot{x}(t).$$

Thus, we can see in Stratonovich calculus, functions obey the same chain rule as in calculus. This is especially useful for stochastic thermodynamics.

Let the thermodynamic state be $z = (q, p, S)$, and the Hamiltonian is $H(q, p, S)$, when applying the chain rule, we will get

$$dH = \partial_q H \circ dq + \partial_p H \circ dp + \partial_S H \circ dS.$$

This formula allows the stochastic first law to be applied by direct substitution of the equations of motion into dH .

The Stratonovich convention also helps in determining the appropriate discretization. Since the Stratonovich integral is defined by midpoint evaluation, a one-step approximation of

$$dx = A(x) dt + B(x) \circ dW$$

will therefore be

$$\begin{aligned} x_{n+1} - x_n &= hA\left(\frac{x_n + x_{n+1}}{2}\right) \\ &+ B\left(\frac{x_n + x_{n+1}}{2}\right) \Delta W_n, \end{aligned}$$

Hence this motivates the use of the Stratonovich convention. It is compatible with midpoint discretization and preserves the ordinary chain rule, and will thus be important for constructing structure preserving schemes later [5].

3 VARIATIONAL PRINCIPLE OF STOCHASTIC THERMODYNAMICS

We start by going over the variational formulation for stochastic thermodynamics. We begin with a simple isolated mechanical system, following the variational formulation of nonequilibrium thermodynamics and its stochastic extension [1, 2, 3].

The thermodynamic phase space is

$$\Omega = T^*Q \times \mathbb{R}, \quad z = (q, p, S) \in \Omega,$$

where q is the position, p is the momentum, and S is the thermodynamic entropy variable. For a simple isolated thermomechanical system with one mechanical degree of freedom, we have the Lagrangian of

$$L(q, v, S) = \frac{1}{2}m|v|^2 - U(q, S),$$

where $U(q, S)$ is the internal energy. The temperature is defined by

$$T(q, S) = \frac{\partial U}{\partial S}(q, S).$$

Therefore, we have

$$\frac{\partial L}{\partial S}(q, v, S) = -\frac{\partial U}{\partial S}(q, S) = -T(q, S).$$

Taking the Legendre transform gives us the Hamiltonian

$$H(q, p, S) = \langle p, v \rangle - L(q, v, S) = \frac{p^2}{2m} + U(q, S).$$

To write the equations as a first-order system, we use the Lagrange-d'Alembert variational principle. We also introduce a thermodynamic displacement Γ , and a medium entropy variable Σ .

$$\delta \int_{t_1}^{t_2} \left(L(q, v, S) + \langle p, \dot{q} - v \rangle + \dot{\Gamma}(S - \Sigma) \right) dt = 0,$$

subject to kinematic and variational constraints for entropy Σ :

$$\begin{aligned} \frac{\partial L}{\partial S} \dot{\Sigma} &= \langle f(q, p, S) + g(q, p, S) \circ \eta(t), \dot{q} \rangle, \\ \frac{\partial L}{\partial S} \delta \Sigma &= \langle f(q, p, S) + g(q, p, S) \circ \eta(t), \delta q \rangle. \end{aligned}$$

The deterministic part $f(q, p, S) = -\gamma(q, p, S) \frac{p}{m}$ represents friction or dissipation, $g(q, p, S)$ represents the stochastic force and $\eta(t)$ is a Gaussian white noise with the properties:

$$\begin{aligned} \langle \eta(t) \rangle &= 0, \\ \langle \eta(t) \eta(s) \rangle &= 2\delta(t - s). \end{aligned}$$

Then, the variational principle gives the following equations of motion:

$$\begin{cases} \dot{q} = \frac{p}{m}, \\ \dot{p} = -\partial_q U(q, S) - \gamma(q, p, S) \frac{p}{m} + g(q, p, S) \circ \eta(t), \\ \dot{S} = -\frac{1}{T(q, S)} \langle -\gamma(q, p, S) \frac{p}{m} + g(q, p, S) \circ \eta(t), \frac{p}{m} \rangle. \end{cases} \quad (3)$$

However, since the Wiener increment appears in both the momentum equation and the entropy equation. The same stochastic force also contributes to the entropy equation through the mechanical power. Therefore, the p -equation and S -equation must be treated jointly as

$$\begin{pmatrix} \dot{p} \\ \dot{S} \end{pmatrix}_{\text{noise}} = \begin{pmatrix} g(q, p, S) \\ -\frac{g(q, p, S)v}{T(q, S)} \end{pmatrix} \circ \eta(t).$$

For thermodynamic consistency, the dissipative force f and the noise amplitude g must satisfy the fluctuation-dissipation relation (FDR).

The FDR balances dissipation and fluctuations required for the stochastic dynamics to have a consistent entropy-production interpretation [1]. For the scalar one-dimensional thermomechanical system, the FDR is

$$-\gamma \frac{p}{m} = -\frac{D}{T} \frac{p}{m} + g \partial_p g - g \partial_S \left(\frac{g}{T} \right) \frac{p}{m}, \quad D = g^2. \quad (4)$$

4 DETERMINISTIC THERMODYNAMIC DISCRETIZATION

A classical Hamiltonian system is described by the equations:

$$\begin{cases} \dot{q} = \frac{\partial H}{\partial p}, \\ \dot{p} = -\frac{\partial H}{\partial q}, \end{cases}$$

where $H(q, p)$ is the Hamiltonian. By defining

$$\mathbf{y} = \begin{pmatrix} q \\ p \end{pmatrix} \quad \text{and} \quad \nabla H(\mathbf{y}) = \begin{pmatrix} \partial_q H \\ \partial_p H \end{pmatrix},$$

the system can be written compactly as

$$\dot{\mathbf{y}} = J \nabla H(\mathbf{y}), \quad \text{with} \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

We can verify that the Hamiltonian can be conserved because

$$\frac{d}{dt} H(\mathbf{y}) = \nabla H(\mathbf{y}) \cdot \dot{\mathbf{y}} = \nabla H(\mathbf{y}) \cdot (J \nabla H(\mathbf{y})) = 0,$$

which follows from the skew-symmetry of J .

4.1 DISCRETE GRADIENT METHOD

To discretize Hamiltonian systems while preserving energy, the Discrete Gradient Method (DGM) can be used [4]. The discretization takes the form:

$$\mathbf{y}_{n+1} = \mathbf{y}_n + h J \bar{\nabla} H(\mathbf{y}_{n+1}, \mathbf{y}_n),$$

where h is the step size and $\bar{\nabla} H$ is a discrete gradient satisfying the conditions:

$$\begin{cases} \bar{\nabla} H(\mathbf{y}_{n+1}, \mathbf{y}_n)^T (\mathbf{y}_{n+1} - \mathbf{y}_n) = H(\mathbf{y}_{n+1}) - H(\mathbf{y}_n), \\ \bar{\nabla} H(\mathbf{y}, \mathbf{y}) = \nabla H(\mathbf{y}). \end{cases}$$

For any discrete gradient, we can check energy is conserved:

$$\begin{aligned} H(\mathbf{y}_{n+1}) - H(\mathbf{y}_n) &= \bar{\nabla} H(\mathbf{y}_{n+1}, \mathbf{y}_n)^T (\mathbf{y}_{n+1} - \mathbf{y}_n) \\ &= \bar{\nabla} H(\mathbf{y}_{n+1}, \mathbf{y}_n)^T (h J \bar{\nabla} H(\mathbf{y}_{n+1}, \mathbf{y}_n)) \\ &= 0. \end{aligned}$$

For the thermodynamic state $z = (q, p, S)$, we will use the Average Vector Field (AVF) discrete gradient. It is defined by [7] as

$$\bar{\nabla} H(z_n, z_{n+1}) = \int_0^1 \nabla H(z_n + \theta(z_{n+1} - z_n)) d\theta.$$

where its components are

$$\begin{aligned} \bar{\nabla}_q H &= \int_0^1 \partial_q H(z_n + \theta(z_{n+1} - z_n)) d\theta, \\ \bar{\nabla}_p H &= \int_0^1 \partial_p H(z_n + \theta(z_{n+1} - z_n)) d\theta, \\ \bar{\nabla}_S H &= \int_0^1 \partial_S H(z_n + \theta(z_{n+1} - z_n)) d\theta. \end{aligned}$$

The average vector field gradient satisfies by the fundamental theorem of calculus:

$$\begin{aligned} H(z_{n+1}) - H(z_n) &= \int_0^1 \frac{d}{d\theta} H(z_n + \theta(z_{n+1} - z_n)) d\theta, \\ &= \int_0^1 \nabla H(z_n + \theta(z_{n+1} - z_n)) \cdot (z_{n+1} - z_n) d\theta, \\ &= \bar{\nabla} H(z_n, z_{n+1}) \cdot (z_{n+1} - z_n). \end{aligned}$$

5 STOCHASTIC THERMODYNAMIC DISCRETIZATION

For a Stratonovich SDE

$$dx = A(x) dt + B(x) \circ dW,$$

the implicit midpoint discretization [5] is

$$x_{n+1} = x_n + A(x_{n+\frac{1}{2}}) \Delta t + B(x_{n+\frac{1}{2}}) \Delta W_n,$$

where

$$x_{n+\frac{1}{2}} = \frac{x_n + x_{n+1}}{2}.$$

We use midpoint evaluation for the Stratonovich noise and an AVF discrete gradient for the Hamiltonian terms. This choice is consistent with the Stratonovich interpretation and simplifies the discrete FDR derivation [5].

Let $\Delta t = h$ and let ΔW_n be the Wiener increment with $\mathbb{E}[\Delta W_n] = 0$, $\mathbb{E}[\Delta W_n^2] = 2h$, consistent with $\langle \eta(t) \eta(t') \rangle = 2\delta(t - t')$.

We define the midpoints

$$\bar{q}_n = \frac{q_n + q_{n+1}}{2}, \bar{p}_n = \frac{p_n + p_{n+1}}{2}, \bar{S}_n = \frac{S_n + S_{n+1}}{2}.$$

such that $\bar{z}_n = (\bar{q}_n, \bar{p}_n, \bar{S}_n)$.

Let

$$\bar{\nabla} H(z_n, z_{n+1}) = (\bar{\nabla}_q H, \bar{\nabla}_p H, \bar{\nabla}_S H)$$

be any discrete gradient satisfying the discrete chain rule above. We will use the average vector field (AVF) discrete gradient

$$\bar{\nabla} H(z_n, z_{n+1}) = \int_0^1 \nabla H(z_n + \theta(z_{n+1} - z_n)) d\theta,$$

due to its high accuracy. Let $z(\theta) = z_n + \theta(z_{n+1} - z_n)$, then each component will be

$$\begin{aligned} \bar{\nabla}_q H &= \int_0^1 \frac{\partial H}{\partial q}(z(\theta)) d\theta, \\ \bar{\nabla}_p H &= \int_0^1 \frac{\partial H}{\partial p}(z(\theta)) d\theta, \\ \bar{\nabla}_S H &= \int_0^1 \frac{\partial H}{\partial S}(z(\theta)) d\theta. \end{aligned}$$

We consider the scheme of a simple adiabatic stochastic system,

$$\begin{cases} q_{n+1} - q_n = h \bar{\nabla}_p H, \\ p_{n+1} - p_n = -h \bar{\nabla}_q H + h f(\bar{z}_n) + g(\bar{z}_n) \Delta W_n, \\ \bar{\nabla}_S H(S_{n+1} - S_n) = -\langle h f(\bar{z}_n) + g(\bar{z}_n) \Delta W_n, \bar{\nabla}_p H \rangle. \end{cases} \quad (5)$$

Applying the AVF into (5), we get

$$\begin{cases} q_{n+1} - q_n = h \int_0^1 \frac{\partial H}{\partial p}(z(\theta)) d\theta, \\ p_{n+1} - p_n = -h \int_0^1 \frac{\partial H}{\partial q}(z(\theta)) d\theta + h f(\bar{z}_n) + \\ g(\bar{z}_n) \Delta W_n, \\ \left(\int_0^1 \frac{\partial H}{\partial S}(z(\theta)) d\theta \right) (S_{n+1} - S_n) = \\ -\langle h f(\bar{z}_n) + g(\bar{z}_n) \Delta W_n, \bar{\nabla}_p H \rangle. \end{cases} \quad (6)$$

First Law. Using discrete chain rule,

$$H(z_{n+1}) - H(z_n) = \bar{\nabla}_q H \cdot \Delta q_n + \bar{\nabla}_p H \cdot \Delta p_n + \bar{\nabla}_S H \Delta S_n,$$

which follows from the discrete gradient property, and using (5), we get

$$\begin{aligned} H(z_{n+1}) - H(z_n) = \\ \bar{\nabla}_q H \cdot h \bar{\nabla}_p H + \bar{\nabla}_p H \cdot (-h \bar{\nabla}_q H + h f(\bar{z}_n) + g(\bar{z}_n) \Delta W_n) \\ - (h f(\bar{z}_n) + g(\bar{z}_n) \Delta W_n) \cdot \bar{\nabla}_p H. \end{aligned}$$

The following two terms cancel:

$$\bar{\nabla}_q H \cdot (h \bar{\nabla}_p H) + \bar{\nabla}_p H \cdot (-h \bar{\nabla}_q H) = 0.$$

As $\Delta q_n = h \bar{\nabla}_p H$, the following terms also cancel each other

$$\begin{aligned} \bar{\nabla}_p H \cdot (h f(\bar{z}_n) + g(\bar{z}_n) \Delta W_n) = \\ \langle h f(\bar{z}_n) + g(\bar{z}_n) \Delta W_n, \bar{\nabla}_p H \rangle. \end{aligned}$$

Therefore,

$$H(z_{n+1}) - H(z_n) = 0.$$

6 STOCHASTIC ENTROPY PRODUCTION

In this section, we will derive the entropy production for the stochastic discrete gradient scheme above. Since entropy production is defined as the irreversibility of a trajectory, we must therefore find the logarithm of the ratio between the probability of a forward path and the probability of the time-reversed path [8, 9]. Therefore, in the discrete case, we must construct the one-step transition kernel and compare it with the kernel of the reversed one-step update.

6.1 COVARIANCE AND CONSTRAINTS

This section derives the one-step transition kernel corresponding to the stochastic discrete-gradient scheme. For simplicity, we will use the one-dimensional scalar-noise case.

$$z_n = (q_n, p_n, S_n), \quad \bar{z}_n = \frac{z_n + z_{n+1}}{2}.$$

We define the following,

$$\begin{aligned} \bar{Q}_n &= \bar{\nabla}_q H(z_n, z_{n+1}), \\ \bar{v}_n &= \bar{\nabla}_p H(z_n, z_{n+1}), \\ \bar{T}_n &= \bar{\nabla}_S H(z_n, z_{n+1}), \\ f_n &= f(\bar{z}_n), g_n = g(\bar{z}_n). \end{aligned}$$

Using our discrete scheme,

$$\begin{cases} \Delta q_n &= h \bar{v}_n, \\ \Delta p_n &= -h \bar{Q}_n + h f_n + g_n \Delta W_n, \\ \bar{T}_n \Delta S_n &= -\bar{v}_n (h f_n + g_n \Delta W_n). \end{cases}$$

If we let a to be,

$$\bar{a}_n = \frac{\Delta p_n}{h} + \bar{Q}_n,$$

it follows that the momentum equation gives us

$$h \bar{a}_n = h f_n + g_n \Delta W_n,$$

Substituting the above into the entropy equation gives us relation

$$\bar{T}_n \Delta S_n = -h \bar{a}_n \bar{v}_n.$$

This shows that the stochastic quantity $h \bar{a}_n$ determines both the momentum increment and the entropy increment.

In the general entropy-dependent case, although the thermodynamic state has three variables $z = (q, p, S)$, there is only one independent Wiener increment ΔW_n . Since the same increment appears in both the momentum equation and the entropy equation, (p, S) -increments cannot be treated as independent Gaussian random variables and must be incorporated into a single transition kernel.

We define

$$\alpha_n = \frac{\bar{v}_n}{\bar{T}_n}.$$

The stochastic residual in the coupled (p, S) -variables is

$$R_n = \begin{pmatrix} \Delta p_n + h \bar{Q}_n - h f_n \\ \Delta S_n + h \alpha_n f_n \end{pmatrix}.$$

Using the numerical scheme, this residual becomes

$$R_n = \begin{pmatrix} g_n \\ -\alpha_n g_n \end{pmatrix} \Delta W_n.$$

Thus

$$R_n = b_n \Delta W_n, \quad b_n = g_n \begin{pmatrix} 1 \\ -\alpha_n \end{pmatrix}.$$

The covariance matrix is

$$D_n = b_n b_n^T = g_n^2 \begin{pmatrix} 1 & -\alpha_n \\ -\alpha_n & \alpha_n^2 \end{pmatrix}.$$

and the one step covariance will be $2h D_n$. This matrix has rank one:

$$\det D_n = 0, \quad \text{rank } D_n = 1.$$

Hence the one-step transition kernel is singular in the full (p, S) -space. Its support is restricted to the range of D_n . Equivalently, the residual must satisfy the constraint

$$\Delta S_n + \alpha_n (\Delta p_n + h \bar{Q}_n) = 0.$$

Using the definition of α_n , this may also be written as

$$\bar{T}_n \Delta S_n + \bar{v}_n (\Delta p_n + h \bar{Q}_n) = 0.$$

The term $\Delta p_n + h\bar{Q}_n$ is the discrete momentum change after subtracting the Hamiltonian force. After multiplying it by the discrete velocity $\bar{v}_n$, we can get the stochastic and dissipative mechanical power over one step. The entropy increment is chosen so that this mechanical power is exactly balanced by the thermal contribution $\bar{T}_n \Delta S_n$. Thus, this support condition ensures the stochastic momentum work is balanced by the entropy increment.

6.2 FORWARD ONE STEP KERNEL

The implicit stochastic update can now be written as a one-step transition probability. The transition kernel will include the Dirac delta functions that impose the deterministic constraints. It will also include the Jacobian factor that accounts for the change of variables from the noise increment w to the endpoint variables. Hence, the transition kernel is not of regular density on the full (q, p, S) -space but a singular measure satisfying the discrete scheme.

For a fixed value w of the Wiener increment, we will define the constraint functions

$$\begin{aligned} G_{q,n}^F(z_{n+1}; z_n, w) &= \Delta q_n - h\bar{v}_n, \\ G_{p,n}^F(z_{n+1}; z_n, w) &= \Delta p_n + h\bar{Q}_n - hf_n - g_n w, \\ G_{S,n}^F(z_{n+1}; z_n, w) &= \Delta S_n + \alpha_n(hf_n + g_n w). \end{aligned}$$

We let

$$\mathcal{J}_n^F(w) = \left| \det \frac{\partial(G_{q,n}^F, G_{p,n}^F, G_{S,n}^F)}{\partial(q_{n+1}, p_{n+1}, S_{n+1})} \right|.$$

This Jacobian contains all endpoint derivatives of $\bar{Q}_n, \bar{v}_n, \bar{T}_n, f_n$ and g_n and is required because the scheme is implicit since the w is mapped to z_{n+1} through nonlinear equations, and thus the transition kernel must include the corresponding change-of-variables factor.

The kernel can be written as the constrained integral

$$K_h^F(z_{n+1} | z_n) = \int_{\mathbb{R}} \rho_h(w) \mathcal{J}_n^F(w) \delta(G_{q,n}^F) \quad (7)$$

$$\times \delta(G_{p,n}^F) \delta(G_{S,n}^F) dw. \quad (8)$$

thus, this clearly shows that the only independent random variable is $w = \Delta W_n$, while the three endpoint variables are constrained by the discrete update.

Solving for $G_{p,n}^F = 0$ gives the forward noise value as

$$w_n^F = \frac{\Delta p_n + h\bar{Q}_n - hf_n}{g_n} = \frac{h(\bar{a}_n - f_n)}{g_n}. \quad (9)$$

Since ΔW_n has density,

$$\rho_h(w) = \frac{1}{\sqrt{4\pi h}} \exp\left(-\frac{w^2}{4h}\right),$$

Since $G_{p,n}^F$ depends linearly on w , we can get

$$\begin{aligned} \delta(G_{p,n}^F) &= \delta(\Delta p_n + h\bar{Q}_n - hf_n - g_n w) \\ &= \frac{1}{|g_n|} \delta(w - w_n^F), \end{aligned}$$

Multiplying (9) by g_n , we can get

$$g_n w_n^F = \Delta p_n + h\bar{Q}_n - hf_n,$$

Therefore, we can obtain

$$hf_n + g_n w_n^F = \Delta p_n + h\bar{Q}_n.$$

Hence this yields the entropy constraint as

$$G_{S,n}^F(z_{n+1}; z_n, w_n^F) = \Delta S_n + \alpha_n(\Delta p_n + h\bar{Q}_n).$$

Therefore, substituting all the constraints and the noise value into (8), the exact local forward kernel is therefore

$$\begin{aligned} K_h^F(z_{n+1} | z_n) &= \frac{\mathcal{J}_n^F(w_n^F)}{|g_n| \sqrt{4\pi h}} \exp\left[-\frac{(w_n^F)^2}{4h}\right] \\ &\times \delta(\Delta q_n - h\bar{v}_n) \\ &\times \delta(\Delta S_n + \alpha_n(\Delta p_n + h\bar{Q}_n)). \quad (10) \end{aligned}$$

6.3 REVERSED ONE STEP KERNEL

To construct the reversed kernel, we impose the following time-reversal assumptions.

$$\Theta(q, p, S) = (q, -p, S).$$

Thus, position and entropy are even variables under time reversal, while momentum is odd. The reversed step is therefore the transition from Θz_{n+1} to Θz_n .

We assume

$$\begin{aligned} H(q, -p, S) &= H(q, p, S), \\ f(q, -p, S) &= -f(q, p, S), \\ g(q, -p, S) &= g(q, p, S) \end{aligned}$$

ensure that the reversed update has the same thermodynamic structure as the forward update

For $f = -\gamma \partial_p H$, the second condition follows if

$$\gamma(q, -p, S) = \gamma(q, p, S).$$

We further require the discrete gradient to be equivariant:

$$\begin{aligned} \bar{\nabla} H(\Theta z_{n+1}, \Theta z_n) &= D\Theta \bar{\nabla} H(z_n, z_{n+1}), \\ D\Theta &= \text{diag}(1, -1, 1). \end{aligned}$$

Consequently,

$$\bar{Q}_n^R = \bar{Q}_n, \quad \bar{v}_n^R = -\bar{v}_n, \quad \bar{T}_n^R = \bar{T}_n,$$

and hence

$$\alpha_n^R = -\alpha_n, \quad f_n^R = -f_n, \quad g_n^R = g_n.$$

For the reversed transition $\Theta z_{n+1} \rightarrow \Theta z_n$, its increments are

$$\Delta q_n^R = -\Delta q_n, \quad \Delta p_n^R = \Delta p_n, \quad \Delta S_n^R = -\Delta S_n.$$

The reversed p -residual is therefore

$$G_{p,n}^R(w) = \Delta p_n + h\bar{Q}_n + hf_n - g_n w.$$

Solving for $G_{p,n}^R(w) = 0$ gives the reversed noise value as

$$w_n^R = \frac{\Delta p_n + h\bar{Q}_n + hf_n}{g_n} = \frac{h(\bar{a}_n + f_n)}{g_n}. \quad (11)$$

The reversed q -constraint satisfies

$$-\Delta q_n + h\bar{v}_n = -(\Delta q_n - h\bar{v}_n),$$

hence

$$\delta(-\Delta q_n + h\bar{v}_n) = \delta(\Delta q_n - h\bar{v}_n).$$

The reversed (p, S) -support constraint is

$$-\Delta S_n - \alpha_n(\Delta p_n + h\bar{Q}_n) = -[\Delta S_n + \alpha_n(\Delta p_n + h\bar{Q}_n)],$$

and therefore

$$\begin{aligned} \delta(-\Delta S_n - \alpha_n(\Delta p_n + h\bar{Q}_n)) &= \\ \delta(\Delta S_n + \alpha_n(\Delta p_n + h\bar{Q}_n)). \end{aligned}$$

Hence the reverse kernel is found to be

$$K_h^R(\Theta z_n | \Theta z_{n+1}) = \frac{\mathcal{J}_n^R(w_n^R)}{|g_n| \sqrt{4\pi h}} \exp\left[-\frac{(w_n^R)^2}{4h}\right] \quad (12)$$

$$\times \delta(-\Delta q_n + h\bar{v}_n) \quad (13)$$

$$\times \delta(-\Delta S_n - \alpha_n(\Delta p_n + h\bar{Q}_n)). \quad (14)$$

6.4 MEDIUM ENTROPY

The medium entropy increment is defined by [1] as the logarithm of the ratio forward and reversed one-step measures:

$$\Delta s_{m,n}^h = \log \frac{K_h^F(z_{n+1} | z_n)}{K_h^R(\Theta z_n | \Theta z_{n+1})}.$$

Because the forward and reversed kernels have the same support constraints, the Dirac deltas cancel in the ratio. The Gaussian normalizations also cancel each other. Therefore, substituting (10) and (14) we will obtain

$$\begin{aligned} \frac{K_h^F(z_{n+1} | z_n)}{K_h^R(\Theta z_n | \Theta z_{n+1})} &= \frac{\mathcal{J}_n^F(w_n^F) \exp[-(w_n^F)^2/(4h)]}{\mathcal{J}_n^R(w_n^R) \exp[-(w_n^R)^2/(4h)]} \\ &= \frac{\mathcal{J}_n^F(w_n^F)}{\mathcal{J}_n^R(w_n^R)} \exp\left[\frac{(w_n^R)^2 - (w_n^F)^2}{4h}\right], \end{aligned}$$

which gives

$$\Delta s_{m,n}^h = \frac{(w_n^R)^2 - (w_n^F)^2}{4h} + \log \frac{\mathcal{J}_n^F(w_n^F)}{\mathcal{J}_n^R(w_n^R)}.$$

To find the first term in the medium entropy equation, we substitute (9) and (11) in and obtain

$$\begin{aligned} \frac{(w_n^R)^2 - (w_n^F)^2}{4h} &= \frac{h}{4g_n^2} [(\bar{a}_n + f_n)^2 - (\bar{a}_n - f_n)^2] \\ &= \frac{h}{4g_n^2} (4\bar{a}_n f_n) \\ &= \frac{h\bar{a}_n f_n}{g_n^2}. \end{aligned} \quad (15)$$

Hence, we can get the medium entropy equation as

$$\Delta s_{m,n}^h = \frac{h\bar{a}_n f_n}{g_n^2} + \log \frac{\mathcal{J}_n^F(w_n^F)}{\mathcal{J}_n^R(w_n^R)}. \quad (16)$$

The second term that we obtained is due to the mid-point update being a nonlinear change of variables from the noise increment to the endpoint z_{n+1} . That endpoint map has a Jacobian, and thus forward and reversed Jacobians do not cancel. We will therefore have to expand the Jacobian ratio for a small h .

6.5 EXPANSION OF THE JACOBIAN RATIO

We will now expand the Jacobian ratio in the discrete entropy increment. Recall that for the general Hamiltonian notation, we use

$$v(z) = \partial_p H(z), \quad T(z) = \partial_S H(z),$$

and introduce the noise vector on the full thermodynamic phase space

$$B(z) = \begin{pmatrix} 0 \\ g(z) \\ -g(z)v/T(z) \end{pmatrix}. \quad (17)$$

Since $\Delta W_n = O_p(h^{1/2})$, and the stochastic part contains $B(z)\Delta W_n$, the stochastic displacement is order $O_p(h^{1/2})$. The deterministic drift part is order $O(h)$, so the stochastic part dominates for small h . Hence the leading size of the one step size is

$$z_{n+1} - z_n = O_p(h^{1/2}).$$

For our AVF discrete gradient,

$$\bar{\nabla} H(z_n, z_{n+1}) = \int_0^1 \nabla H(z_n + \theta \Delta z_n) d\theta.$$

Because the AVF average is symmetric around the midpoint, the first-order error cancels. Therefore

$$\bar{\nabla} H(z_n, z_{n+1}) = \nabla H(\bar{z}_n) + O_p(h).$$

We then differentiate the AVF discrete gradient with respect to the endpoint z_{n+1} . Since

$$\frac{\partial}{\partial z_{n+1}} (z_n + \theta(z_{n+1} - z_n)) = \theta I,$$

we obtain the differential under the integral as

$$D_2 \bar{\nabla} H(z_n, z_{n+1}) = \int_0^1 \theta \nabla^2 H(z_n + \theta \Delta z_n) d\theta.$$

For small Δz_n ,

$$\nabla^2 H(z_n + \theta \Delta z_n) = \nabla^2 H(\bar{z}_n) + O_p(h^{1/2}),$$

this yields

$$D_2 \bar{\nabla} H(z_n, z_{n+1}) = \frac{1}{2} \nabla^2 H(\bar{z}_n) + O_p(h^{1/2}).$$

Similarly, for the discrete noise vector,

$$D_2 \mathbf{B}_n^h = \frac{1}{2} D\mathbf{B}(\bar{z}_n) + O_p(h^{1/2}). \quad (18)$$

The forward constraint map is

$$G_n^F(z_{n+1}, w) = z_{n+1} - z_n - hA_n^h - w\mathbf{B}_n^h.$$

where A_n^h denotes the deterministic discrete drift and $\mathbf{B}_n^h$ denotes the discrete noise vector.

Differentiating with respect to z_{n+1} , we will obtain

$$D_2 G_n^F = I - hD_2 A_n^h - wD_2 \mathbf{B}_n^h.$$

Since the deterministic increment is $hD_2 A_n^h = O(h)$ and the stochastic increment is $wD_2 \mathbf{B}_n^h = O_p(h^{1/2})$, the leading nontrivial contribution to the Jacobian comes from the noise term. Therefore, we will get

$$D_2 G_n^F = I - w_n^F D_2 \mathbf{B}_n^h + O_p(h).$$

Substituting (18) into the above, we can obtain

$$D_2 G_n^F = I - \frac{w_n^F}{2} D\mathbf{B}(\bar{z}_n) + O_p(h).$$

Using the log-det identity,

$$\log \det(I + M) = \text{tr } M + O(\|M\|^2).$$

and

$$M = -\frac{w_n^F}{2} D\mathbf{B}(\bar{z}_n) + O_p(h)$$

Since $w_n^F = O_p(h^{1/2})$, we have $\|M\|^2 = O_p(h)$. Therefore

$$\log \mathcal{J}_n^F(w_n^F) = -\frac{w_n^F}{2} \text{tr } D\mathbf{B}(\bar{z}_n) + O_p(h)$$

Since $\text{tr } D\mathbf{B} = \nabla_z \cdot \mathbf{B}$, we will get

$$\log \mathcal{J}_n^F(w_n^F) = -\frac{w_n^F}{2} \nabla_z \cdot \mathbf{B}(\bar{z}_n) + O_p(h). \quad (19)$$

Under the time-reversal assumptions above, $v = \partial_p H$ is odd, $T = \partial_S H$ is even, and g is even. Hence the noise vector satisfies

$$\mathbf{B}(\Theta z) = -D\Theta \mathbf{B}(z).$$

Differentiating the above, we will get

$$D\mathbf{B}(\Theta z) D\Theta = -D\Theta D\mathbf{B}(z).$$

Since $(D\Theta)^2 = I$, this implies that

$$D\mathbf{B}(\Theta z) = -D\Theta D\mathbf{B}(z) D\Theta.$$

Taking traces and using cyclicity of the trace,

$$\begin{aligned} \nabla_z \cdot \mathbf{B}(\Theta z) &= -\text{tr}(D\Theta D\mathbf{B}(z) D\Theta) \\ &= -\text{tr}(D\mathbf{B}(z) (D\Theta)^2) \\ &= -\nabla_z \cdot \mathbf{B}(z). \end{aligned} \quad (20)$$

Applying the logarithmic determinant expansion, this produces

$$\log \mathcal{J}_n^R(w_n^R) = \frac{w_n^R}{2} \nabla_z \cdot \mathbf{B}(\bar{z}_n) + O_p(h). \quad (21)$$

Subtracting (21) from (19) gives

$$\begin{aligned} \log \frac{\mathcal{J}_n^F(w_n^F)}{\mathcal{J}_n^R(w_n^R)} &= -\frac{w_n^F + w_n^R}{2} \nabla_z \cdot \mathbf{B}(\bar{z}_n) + O_p(h), \\ &= -\frac{h\bar{a}_n}{g_n} \nabla_z \cdot \mathbf{B}(\bar{z}_n) + O_p(h). \end{aligned} \quad (22)$$

Because the q -component of $\mathbf{B}$ is zero, we get the divergence of $\mathbf{B}$ as

$$\begin{aligned} \nabla_z \cdot \mathbf{B} &= \partial_q(0) + \partial_p g + \partial_S \left(-\frac{gv}{T} \right) \\ &= \partial_p g - \partial_S \left(\frac{gv}{T} \right). \end{aligned} \quad (23)$$

Substituting (22) into the medium entropy equation, we get

$$\begin{aligned} \Delta s_{m,n}^h &= \frac{h\bar{a}_n f_n}{g_n^2} + \log \frac{\mathcal{J}_n^F(w_n^F)}{\mathcal{J}_n^R(w_n^R)} \\ &= \frac{h\bar{a}_n}{g_n^2} [f_n - g_n \nabla_z \cdot \mathbf{B}(\bar{z}_n)] + O_p(h). \end{aligned} \quad (24)$$

Substituting (23) into the medium entropy equation (24), we can get the medium entropy as

$$\Delta s_{m,n}^h = \frac{h\bar{a}_n}{g_n^2} \left[f_n - g_n \left(\partial_p g - \partial_S \left(\frac{gv}{T} \right) \right)_{\bar{z}_n} \right] + O_p(h). \quad (25)$$

For a general Hamiltonian, the S -derivative expands to

$$\partial_S \left(\frac{gv}{T} \right) = v \partial_S \left(\frac{g}{T} \right) + \frac{g}{T} \partial_S v,$$

Multiplying (23) by g and using the above S -derivative, we will obtain

$$g \nabla_z \cdot \mathbf{B} = g \partial_p g - gv \partial_S \left(\frac{g}{T} \right) - \frac{g^2}{T} \partial_S v. \quad (26)$$

Substituting (26) into (24), the medium entropy equation can be equivalently written as

$$\Delta s_{m,n}^h = \frac{h\bar{a}_n}{g_n^2} \left[f_n - \left(g \partial_p g - g v \partial_S \left(\frac{g}{T} \right) - \frac{g^2}{T} \partial_S v \right) \right] + O_p(h).$$

6.6 GENERAL FDR

Thermodynamic consistency is imposed by matching the thermodynamic entropy increment with the path-probability medium entropy. From the discrete thermodynamic entropy equation,

$$\bar{T}_n \Delta S_n = -\bar{v}_n (h f_n + g_n \Delta W_n).$$

Using

$$h\bar{a}_n = h f_n + g_n \Delta W_n,$$

we obtain the exact identity

$$\Delta S_n = -\frac{h\bar{a}_n \bar{v}_n}{\bar{T}_n}.$$

To make the path-probability medium entropy agree with the thermodynamic entropy increment, we impose the following to derive a leading-order discrete FDR

$$\Delta S_n = \Delta s_{m,n}^h + O_p(h).$$

We match the thermodynamic entropy increment to the medium entropy equation, therefore we require

$$\frac{h\bar{a}_n}{g_n^2} [f_n - g_n \nabla_z \cdot \mathbf{B}(\bar{z}_n)] + O_p(h) = -\frac{h\bar{a}_n \bar{v}_n}{\bar{T}_n}.$$

Since the equality is required to hold for arbitrary realizations of the leading stochastic increment $h\bar{a}_n = O_p(h^{1/2})$, the coefficients of this leading term must agree.

Matching the coefficient of the leading stochastic increment $h\bar{a}_n = O_p(h^{1/2})$, we obtain

$$\frac{f_n - g_n \nabla_z \cdot \mathbf{B}(\bar{z}_n)}{g_n^2} = -\frac{\bar{v}_n}{\bar{T}_n}.$$

Therefore,

$$f_n = -\frac{g_n^2}{\bar{T}_n} \bar{v}_n + g_n \nabla_z \cdot \mathbf{B}(\bar{z}_n).$$

Using

$$\nabla_z \cdot \mathbf{B} = \partial_p g - \partial_S \left(\frac{g v}{T} \right),$$

the general entropy-dependent discrete FDR is

$$f(\bar{z}_n) = -\frac{g(\bar{z}_n)^2}{\bar{T}_n} \bar{v}_n + g(\bar{z}_n) \partial_p g(\bar{z}_n) - g(\bar{z}_n) \partial_S \left(\frac{g v}{T} \right) (\bar{z}_n).$$

Equivalently, expanding the S -derivative gives

$$\begin{aligned} f(\bar{z}_n) = & -\frac{g(\bar{z}_n)^2}{\bar{T}_n} \bar{v}_n + g(\bar{z}_n) \partial_p g(\bar{z}_n) \\ & - g(\bar{z}_n) v(\bar{z}_n) \partial_S \left(\frac{g}{T} \right) (\bar{z}_n) - \frac{g(\bar{z}_n)^2}{T(\bar{z}_n)} \partial_S v(\bar{z}_n). \end{aligned}$$

This is the general form. The final term must be kept unless $v = \partial_p H$ is independent of S .

To recover the mechanical system used in Section 3, we now impose the additional mechanical Hamiltonian assumption

$$H(q, p, S) = \frac{p^2}{2m} + U(q, S).$$

Since the $U(q, S)$ has no p -dependence, this means v depends only on p , not on S . Therefore, the term $(g^2/T) \partial_S v$ vanishes. The FDR then reduces to

$$\begin{aligned} f(\bar{z}_n) = & -\frac{g(\bar{z}_n)^2}{\bar{T}_n} \bar{v}_n + g(\bar{z}_n) \partial_p g(\bar{z}_n) \\ & - g(\bar{z}_n) v(\bar{z}_n) \partial_S \left(\frac{g}{T} \right) (\bar{z}_n). \end{aligned}$$

7 CONCLUSION

In summary, this paper reviews how we can construct schemes that respect the thermodynamic structure of the continuous model. For the deterministic case, we reviewed discrete-gradient methods and showed how they can be used to preserve the first law of thermodynamics. We further extended this idea to the stochastic thermodynamic case, by combining discrete-gradient methods with midpoint-type Stratonovich discretization, and proposing a stochastic discrete-gradient scheme. Using the discrete chain rule, we also showed that the proposed scheme satisfied the first law. We further formulated the entropy-production framework in accordance with the Fluctuation-Dissipation relation in order to show it follows the second law. We ultimately derived an entropy-dependent discrete fluctuation-dissipation relation.

Further work includes testing the proposed scheme using concrete examples. While the present work provides a theoretical foundation for thermodynamically consistent stochastic discretization, next steps are numerical simulations to test whether the proposed discretization remains consistent and whether there is long-time stability and convergence. We can also further extend the present finite-dimensional formulation to more complex systems.

ACKNOWLEDGMENTS

I would like to thank Professor François Gay-Balmaz for his invaluable guidance and support throughout this project. I would like to thank Hector Vaquero del Pino for his assistance throughout this project.

I would also like to acknowledge the funding support from Nanyang Technological University – URECA Undergraduate Research Programme for this research project.

REFERENCES

- [1] Héctor Vaquero del Pino, François Gay-Balmaz, Hiroaki Yoshimura, and Lock Yue Chew. A variational formulation of stochastic thermodynamics. part i: Finite-dimensional systems. *arXiv preprint arXiv:2510.01787*, 2025.
- [2] François Gay-Balmaz and Hiroaki Yoshimura. From lagrangian mechanics to nonequilibrium thermodynamics: A variational perspective. *Entropy*, 21(1):8, 2018.
- [3] François Gay-Balmaz and Hiroaki Yoshimura. A variational formulation of nonequilibrium thermodynamics for discrete open systems with mass and heat transfer. *Entropy*, 20(3):163, 2018.
- [4] Oscar Gonzalez. Time integration and discrete hamiltonian systems. *Journal of Nonlinear Science*, 6(5):449–467, 1996.
- [5] Peter E. Kloeden and Eckhard Platen. *Numerical Solution of Stochastic Differential Equations*. Springer, 1992.
- [6] Bernt Oksendal. *Stochastic differential equations: an introduction with applications*. Springer Science & Business Media, 2013.
- [7] G. R. W. Quispel and D. I. McLaren. A new class of energy-preserving numerical integration methods. *Journal of Physics A: Mathematical and Theoretical*, 41:045206, 2008.
- [8] Udo Seifert. Entropy production along a stochastic trajectory and an integral fluctuation theorem. *Physical review letters*, 95(4):040602, 2005.
- [9] Udo Seifert. Stochastic thermodynamics, fluctuation theorems and molecular machines. *Reports on progress in physics*, 75(12):126001, 2012.